

# Intermediate Macroeconomics

## Lecture 1

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# Outline

- 1 Motivation
  - More than the sum of its parts
  - Some Australian Macro data
- 2 Practical info
- 3 Class Structure
- 4 Class Objectives
  - Economic Well-being - Concepts and measurement
  - Schools of Thought
  - Income distribution
- 5 Math Basics



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# Recent macroeconomic headlines

- “Weak consumer confidence shows rate cut still needed”
- “Jobless rate jumps...”
- “RBA leaves interest rates on hold”
- “RBA: It is time to rethink bank card surcharges”
- “Australian dollar hits nine-month high against US dollar”
- “Trump trade war blamed for surprise shrinking of US economy”
- “Australia’s demand for power has surged...”
- “Drought and soaring water prices push wine grape growers...”



# The challenge

Popular media often muddles the causes and effects of economic phenomena. The challenge is...

- Differentiating which phenomena are the events that drive economic outcomes and which phenomena are the consequence of these drivers.

The remedy is...

- Impose structure on economic thinking through the use of mathematically derived models.



# Macroeconomics is...

- The macro-economy is the product of millions of individual decisions
  - Households decide
    - what to consume
    - how much to save
    - how much to work
  - Firms decide
    - how much to produce
    - how many workers to hire
    - how much capital (machinery) to employ
  - Decisions are implemented in markets, where prices are set
- Further actions taken by the government and central bank



# Macroeconomics is...

- Understanding individual decisions only partially explain macroeconomic phenomenon
  - output growth
  - business cycle fluctuations
  - the price level and inflation
  - trade
- We need to develop dedicated macroeconomic models



# Some Australian Macro Data

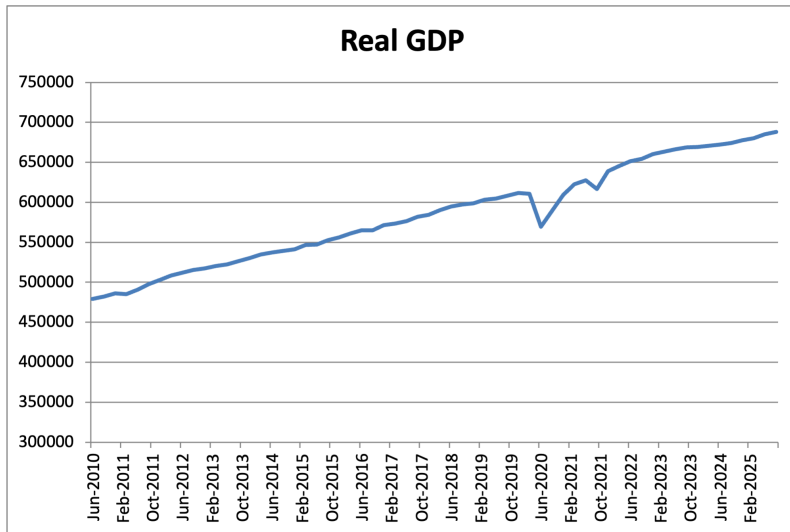


Figure 1: Australian Real GDP



# Some Australian Macro Data

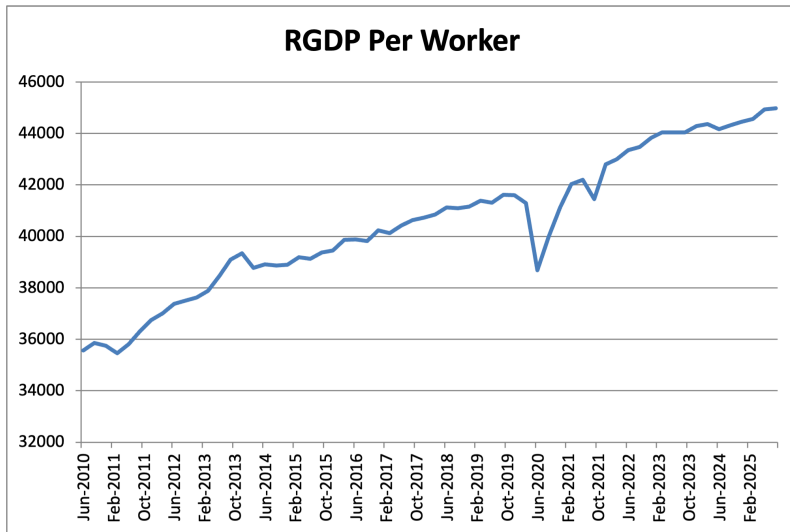


Figure 2: Australian Real GDP per capita



# Some Australian Macro Data

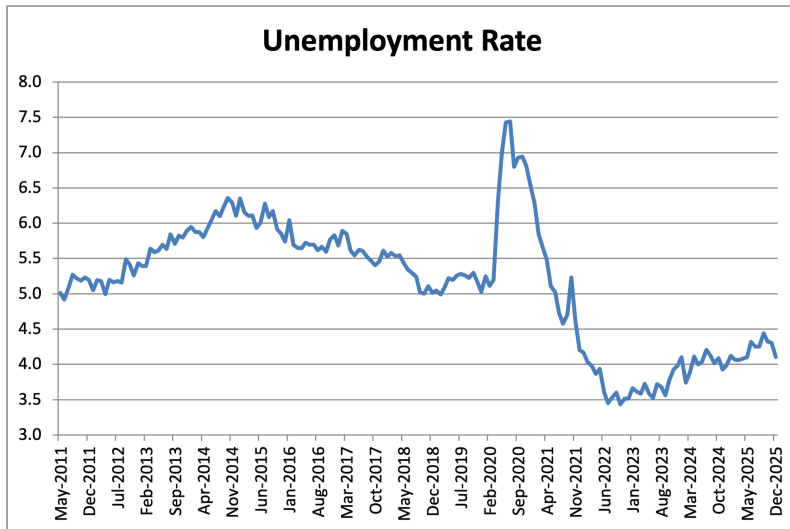


Figure 3: Australian Unemployment Rate



# Some Australian Macro Data

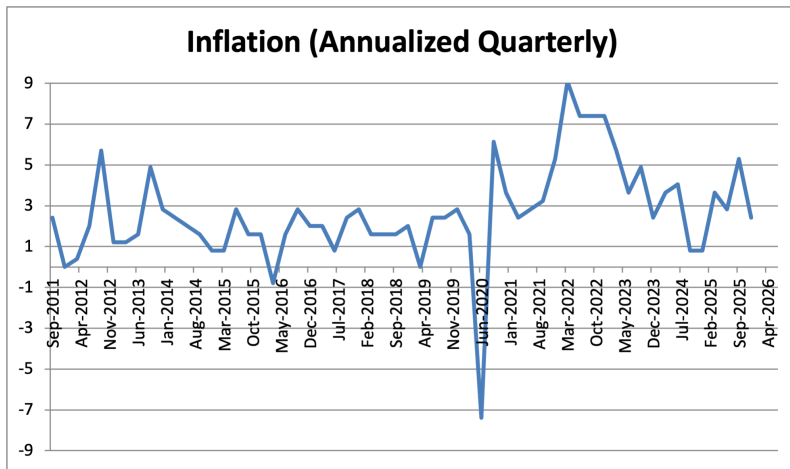


Figure 4: Australian Inflation rate



# Some Australian Macro Data

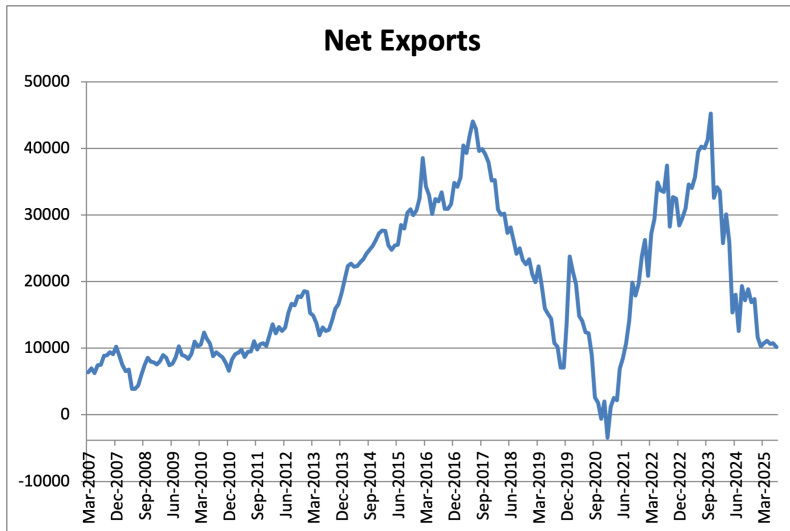


Figure 5: Australian Net Exports



# Some Australian Macro Data

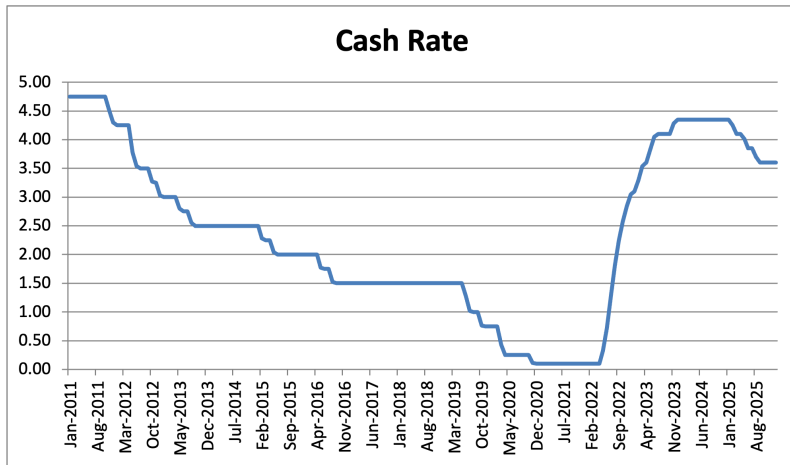


Figure 6: Australian Cash Rate



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  - consultation by appointment



- Skills Development (0%)
  - Each week a set of questions will be made available through Canvas. Answer all the questions to the best of your ability.
  - Answers will be posted after the material has been covered in the following week's tutorial.
- Individual Assignments (40%)
  - Four quizzes during the term to be completed on-line.
  - Will consists of multiple choice, problem solving, and short answer questions.
- Final exam (60%)
  - The cumulative exam covers material throughout the term.
  - Will consists of multiple choice, problem solving, and short answer questions.



# Some tips

- This subject requires continuous effort because each topic builds on the previous ones.
- In order to do well in this subject, each week:
  - ① attend class,
  - ② study the theory explained in class,
  - ③ attempt the tutorial exercises,
  - ④ attend tutorials,
  - ⑤ after tutorial, repeat 2 and 3.



# Some evidence

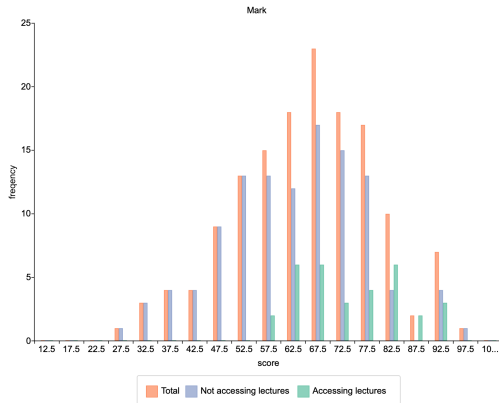


Figure 7: 2024 Autumn mark distribution

- Attended or watched at least 8 lectures: 100% pass rate
- Attended or watched fewer than 8 lectures: 82% pass rate



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- ① Production and Growth
  - Sources of Growth
  - Solow (Classic) growth model to explain the growth process and the determinants of potential output
- ② Demand driven models of Business Cycles
- ③ Resource constraints
  - Labour markets
  - AD-AS
- ④ Micro-foundations
  - Consumption and Investment decisions
  - Labor market: Phillips Curve, Market Frictions
- ⑤ Macroeconomic topics
  - Big Events and Issues
  - Contemporary macroeconomic theory and models



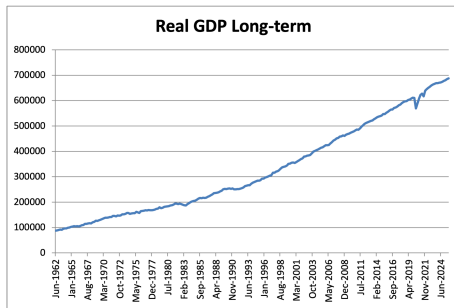


Figure 8: Aust RGDP

Observe: Generally increasing trend.

So ask

- What are the sources of growth?
- Can we model growth so that we can understand difference across time and across countries?



# Business Cycles

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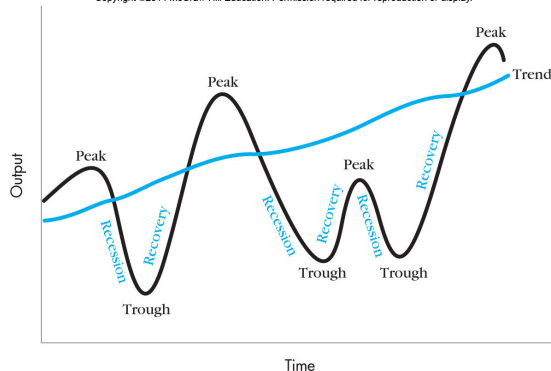


Figure 9: Business Cycles around a trend

## Observe

- Around a trend the economy
  - experiences strong growth during a recovery/boom
  - attains a peak
  - experiences a period of declining output
  - attains a trough
  - recovers with strong growth



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# Class Objectives

- Develop formal **models** with which to understand the macroeconomy
  - Economic growth
  - Business Cycle fluctuations
  - Relations between macroeconomic variables (GDP, Unemployment, Inflation,...)
- Understand different schools of thought
  - Classical
  - Keynesian
  - Monetarists
  - Modern thinking incorporates aspects of these
- Develop the skills to map events into a model
- Use developed models to inform macroeconomic policy



Our interest is in economic well-being. Indicators include:

- Total output (GDP)
- Inflation
- Unemployment
- Income distribution (**only recently included**)



# Gross Domestic Product

**GDP:** final goods and services produced and consumed by the population in a given period.

Three methods of measuring aggregate economic output

- **Production approach:** The sum of the value added at each stage of production for everything domestically produced.
- **Expenditure approach:** A measure of all final Goods and Services sold with appropriate accounting for investment, imports, exports, and government spending
  - $$Y = C + I + G + \underbrace{X - Im}_{=NX}$$
- **Income approach:** A measure of all income earned for domestically performed work and for non-work income earned on domestic assets.



$$Y = C + I + G + NX \quad (1)$$

**Output (Y)** is the sum of the different sources of spending of domestic production

- **Consumption (C)**: Consumer spending on final Goods and Services. The consumer is the final user of the product. Business purchases of inputs to production are captured in the price of the final good
- **Investment (I)**
  - Business purchases of capital: The purchase of an asset is captured in the price of the final good it is used to produce.
  - business changes in inventory: To capture the value of production that spans a year (or whatever time unit employed)
  - Purchase of a new-construction home: Counted as investment rather than consumption because of the long life as a durable asset.



$$Y = C + I + G + NX$$

- **Government Spending (G)**: Government spending on goods and services.
  - The government provides services to the public (law enforcement, national defense, infrastructure maintenance, education, etc.).
  - These services are valued at cost (of buying resources and paying salaries).
  - G does not include transfer payments (benefits paid out to the population).
- **Net Export (NX = Exports - Imports)**
  - Exports: The value of products produced in Australia and exported are counted towards domestic output. Imports sold domestically are included in the total value of Goods and Services sold in Australia.
  - Imports: The value of goods when entering the country needs to be deducted as it is not part of domestic output.



# Other NIPA identities

## Other useful National Income and Product Accounts (NIPA) identities

- Disposable Income: Where it comes from

$$YD = Y + TR - TA \quad (2)$$

- Disposable Income: Where it goes

$$YD = C + S \quad (3)$$

- Equations (1), (2), and (3) can be used to obtain

$$S - I = \underbrace{(G + TR - TA)}_{\text{Budget Deficit}} + \underbrace{NX}_{\text{Trade Surplus}} \quad (4)$$

or

$$S = (G + TR - TA) + I + NX$$

- Savings pays for the government deficit, investment, and net exports.
- With no public or international sectors (4) simplifies to

$$S = I \quad (5)$$



- We measure GDP as the value of output using current dollars.

$$(\text{Nominal})GDP_t = \sum_i p_{it}y_{it}$$

$p_i$  is the price of a product  $y_i$ . From one year to the next, there are two components which contribute to the changes GDP:

- Changes in output ( $Y$ ). What we would like to measure since this represents a real increase in the level of production.
- Changes in prices (inflation). This, we would like to remove from the measure of the change in GDP.



Measures of the rate of change in prices over time.

- **Fix-weighted** price index - the Consumer Price Index (CPI):
  - The government samples consumers to determine typical consumer purchases (the “basket”).
  - Traditionally, once established, the basket remains fixed. The modern method is to “chain-weight” and allow the basket to evolve along with changes in consumption spending.
  - Each year, operatives shop to determine how much the total basket costs. The change in the price of the basket is the measure of inflation.
  - Index is set to 100 in the base year. Increases and decreases in the cost of the basket are reflected in the index.
- **Variable-weighted** price index - the GDP deflator.



Inflation is measured as the growth in the price level (using either the CPI or the GDP deflator)<sup>1</sup>

$$\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t} = \frac{\Delta P_{t+1}}{P_t}$$

- Nominal GDP to Real GDP (RGDP) in base year prices

$$\text{RealGDP}_t = \text{NominalGDP}_t * (P_{base}^{CPI} / P_t^{CPI})$$

---

<sup>1</sup>This “growth” equation works for computing the growth rate of any variable,

$$g_x = \frac{x_{t+1} - x_t}{x_t}$$



Inflation can be a concern (not just a tool to compute real output)

- People and institutions can typically adapt to reasonable levels of anticipated inflation to neutralize the impact of the price changes
- Unanticipated inflation can be harmful in that it redistributes income and wealth.
  - This creates hardship for people and makes planning difficult
- Hyperinflation can be a serious problem as the institutions and accommodations become costly

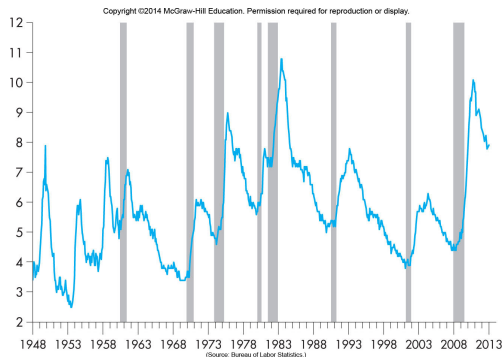


Unemployment has a social and economic cost

- Unemployment has negative consequences on single persons (loss of skills and self-esteem) and society (transfers and health expenditure, social tension);
- High unemployment indicates a breakdown in markets, and a wasting of resources:
  - If there are people who desire work, and who are capable of work, but are unable to find it, there is some breakdown in the economy. It also tends to be associated with diminished levels of output.



# Unemployment



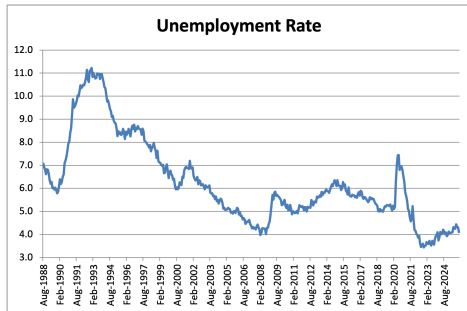
## Observe

- The unemployment rate spikes up during a recession
- Unemployment falls more slowly during the recovery and continued economic growth
- After the 1990-91 and 2001-02 recessions, unemployment continued to rise during the early stages of recovery.

Figure 10: US Unemployment (recessions in gray)



# Unemployment



Observe

- Similar patterns in Australian data
- 2012-2014 rise is an exception

Figure 11: Australian Unemployment



# Unemployment

Survey the population 16 and over whether they are:

- **employed**: paid employment during the last week, even if only for 1 hour
- **unemployed**: not employed, but actively looking for work during the last week
- **not in the labor force**: did not work, not looking
  - Examples, full time students, incarcerated, in the military, stay-at-home parent

The unemployment rate (UR) is calculated as:

$$UR = \frac{\text{unemployed}}{\text{employed} + \text{unemployed}}$$



# Schools of Thought: Classical

Aggregate output is determined by the production resources available to the economy

- Competitive market ensure the efficient allocation of resources to maximize the value of output (markets always in equilibrium)
  - Resources: labour, capital, raw materials, technology
- An economy grows as more resources become available
- A reasonable description of what an economy is capable.
  - There are reasons to be concerned that the economy typically falls short

Championed by Adam Smith

- **New Classical** builds on the notion of the macroeconomy as the aggregation of individual optimizing market-driven activities.
  - Proponents include Robert Barro, Neil Wallace, **Thomas Sargent**, **Robert Lucas**, **Finn Kydland** and **Edward Prescott**
- **Monetarist**: fluctuations are due to mismanagement of money supply, better to leave the economy alone (**Milton Friedman**)



# Schools of Thought: Keynesian

John Maynard Keynes recognized the presence of market inefficiencies, meaning that:

- Resources are not used efficiently
- Output is below what the economy is capable of
- Unemployment is high

Two main fields:

- **Post Keynesian:**
  - recognises that the economy is demand driven and normally not in equilibrium. Also, the distribution of income matters.
  - Proponents include Nicholas Kaldor, Michal Kalecki, Joan Robinson, Paul Davidson, Lance Taylor, Wynne Godley and Marc Lavoie.
- **New Keynesian:**
  - incorporate inefficient markets into supply-driven models of the macroeconomy
  - Proponents include Ben Bernanke, Olivier Blanchard, Gregory Mankiw, Larry Summers, Janet Yellen, George Akerlof and Joseph Stiglitz



Who benefits from growth?

- **Functional distribution of income:** distribution of income between capital and labor;
- **Personal distribution of income:** distribution of income among individuals.



# Functional income distribution

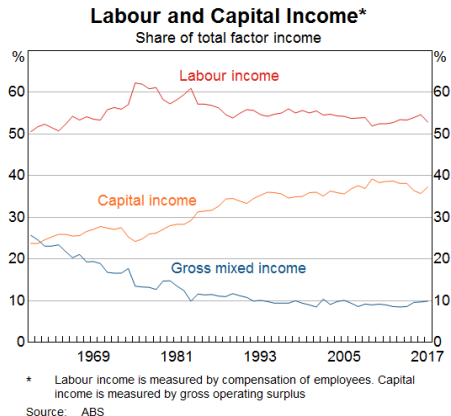


Figure 12: Shares of income in Australia (Source: RBA).



# Functional income distribution



**Figure 13:** Evolution of average wages and labour productivity in selected advanced G20 economies, 2000-2014. Source: ILO Global Wage Database; ILO Trends Econometric Models, July 2015.



# Personal income distribution

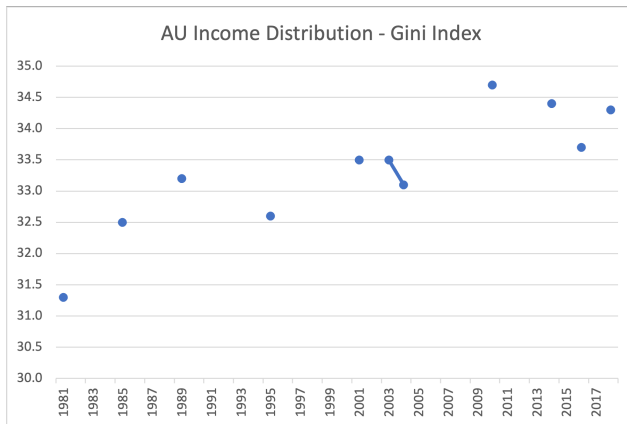


Figure 14: Gini Index, Australian Income Distribution (Source: World Bank)



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# Some Useful Math

Use mathematical principles for upcoming lectures

- Math involving exponents:

$$X^a X^b = X^{a+b}$$

$$X^a / X^b = X^{a-b}$$

$$(X^a)^b = X^{ab}$$

- Logarithmic functions

$$\frac{\partial \ln x}{\partial x} = \frac{1}{x}$$



- Growth Rates:

$$g_X = \frac{\Delta X}{X} = \frac{X_{t+1} - X_t}{X_t}$$

Suppose  $Z = X^a Y^b$  then

$$g_Z = a \cdot g_X + b \cdot g_Y$$

Small differences in the growth rate have substantial impact on GDP over time

- Rule of 70
- An economy growing at  $g\%$  p.a. will double GDP per capita in  $70/g$  years.
  - A country growing at 4% p.a. will double its per capita GDP in 17.5 years.
  - A country growing at 3% p.a. will double its per capita GDP in 23.3 years.

